

ASHISH VERMA

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PROFESSIONAL SUMMARY

M.Tech Computer Science candidate at IIT Guwahati and highly capable developer with a strong foundation in C++. Extensive hands-on experience building full-stack web applications and developing machine learning models, backed by rigorous academic research in advanced algorithms. Passionate about leveraging strong analytical skills and competitive programming expertise to architect highly efficient, scalable software solutions.

EDUCATION

Indian Institute of Technology Guwahati (IITG) Master of Technology in Computer Science and Engineering	2024 - 2026 CGPA: 7.94
Kamla Nehru Institute of Technology Sultanpur (KNIT) Bachelor of Technology in Computer Science and Engineering	2018 - 2022 CGPA: 8.36

PROJECTS

- A Parallel Algorithm for the LCS Problem on PEM Model** | MTech Thesis Project *July 2025 – June 2026*
- **Architected** an I/O-efficient parallel algorithm for the Longest Common Subsequence (LCS) problem, significantly reducing disk-transfer bottlenecks for large-scale datasets.
 - **Developed** a "Tiled Wavefront" scheduling mechanism using the Parallel External Memory (PEM) model to optimize multi-core processor utilization and hide hard disk latency.
 - **Engineered** a conflict-free memory access strategy adhering to CREW constraints, enabling simultaneous matrix tile computations across multiple processors.
 - **Evaluated** the algorithm's theoretical performance using Brent's Theorem, establishing an optimal upper bound for parallel I/O complexity.
- Logic Synthesis Using Heuristics** | 🧠 *Feb. 2025 – Apr. 2025*
- **Designed** a neural network combining **GCN and embedders** to predict and optimize circuit delay in different scaled AIGs, cutting logic synthesis **time from 3 hours (manual) to 20 minutes** with **±4.02% error**.
 - **Tech stack:** Python, PyTorch, Shell Scripts, ABC tool, Few-Shot Learning, Simulated Annealing, Monte Carlo Optimization.
- Multi-Agent PDF Chatbot (RAG)** | 🧠 *Aug. 2025 – Sept. 2025*
- **Engineered** a **Retrieval-Augmented Generation (RAG)** system enabling interactive, persistent, and multi-agent conversations with PDF documents, featuring source-cited responses for verifiable and reliable AI interactions.
 - **Implemented** a scalable architecture using **ChromaDB** for persistent vector storage and **LangChain** for efficient text chunking and retrieval, allowing seamless agent creation and context management.
 - **Tech Stack:** Streamlit, Google Gemini 1.5 Flash, ChromaDB, LangChain, PyMuPDF, Sentence-Transformers.
- Blog-Bar - Full-Stack Blog Platform** | 🧠 *Sept. 2025*
- A **Full-stack** blogging platform designed for creating, sharing, and managing articles. It provides a secure environment for authors to publish content and a **dynamic, searchable** experience for readers.
 - Engineered a secure **Node.js** API for content management and designed a responsive **React** front-end with features like a rich text editor and an admin dashboard.
 - **Tech Stack:** React, Redux, Node.js, Express.js, MongoDB, JWT, and Tailwind CSS.

ACHIEVEMENTS

- **AIR 733**, GATE CSE (**99.40 %ile**) among **123,967 students**
- **Competitive Programming**, Codeforces **Specialist** with a contest rating of **1411**
- **Coding**, Solved **900+** **Leetcode** Problems with rating **1893(Knight)** honing Data Structures and Algorithms(DSA)

SKILLS

- **Languages:** C/C++, Python
- **Web Technologies:** HTML, CSS, JavaScript, React, Node.js, Express.js
- **Databases:** MongoDB, MySQL
- **Tools:** Git, GitHub, Linux (Elementary), Google Colab, Visual Studio, Windows

RELEVANT COURSEWORK

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|----------------------------------|------------------------------|------------------------------|
| • Design & Analysis of Algorithm | • Introduction to ML (NPTEL) | • High Performance Computing |
| • FM for System Verification | • ML for EDA | • Cloud Computing (NPTEL) |
| • Speech Processing | • Data Structures Lab | • Systems Lab |